AT2 Assignment – Real Estate Agency X – Image Classification

Assumptions: -

- The company is a large-sized company employing 120 full-time employees across major cities in the USA. The annual revenue in the 2022 financial year is approximately $1.50 billion. The company has a headquarter in New York and branches in Los Angeles, Chicago, and Houston.
- This project proposal will be presented to the senior management, such as the Head of Operations, Chief Financial Officer, Chief Operating Officer, and the Chief Executive Officer.
- ABC Consultancy is a leading IT consultancy company based in New York. The management has reached out to ABC Consultancy to assist with finding an automated image classification using artificial intelligence techniques to help the company solve this problem.
- The company staff has to manually upload 50,000 new properties each month, and each property has approximately 6 to 10 pictures. The average hourly rate for full-time staff in manual photo categorisation is $30 per hour (or $59,280 per year). The company requires this position for each branch (4 positions in total) due to the volume of work.
- Due to the limitations of words, complete information will be added to the Appendices at the end of the proposal.

A. The current business problems or challenges and its business impacts.

The process of uploading property photos into designated categories is time-consuming and laborious for staff. This manual method consumes time, dissatisfies employees, and increases human resource costs. Therefore, the business is experiencing many adverse effects.

These effects consist of the following: -

1. **Reduced Efficiency**: Manually uploading a lot of property images (over 400,000 pictures) per month is time-consuming, and time spent about 20 minutes per property is a lot of time and work for the staff. This inefficiency slows the company's capacity to market and promote new properties, thereby delaying sales and missing sales prospects.
2. **Increased Operational Costs**: Due to the labour-intensive nature of the procedure, a substantial investment in human resources is necessary. In addition, the company must assign a number of employees to this task, leading to an increase in operating expenses ($59,280 * 4 = $237,120 per year). These expenses could be better allocated to activities that generate revenue or employee development.
3. **Employee Dissatisfaction**: Due to the task's repetitious and monotonous nature, employees responsible for uploading property images have already experienced frustration and dissatisfaction. This can decrease employee morale and productivity and possibly increase turnover rates in the near future. Therefore, maintaining a positive work environment and retaining valuable talent is contingent upon employee satisfaction.
4. **Missed Advertising Potential:** The manual procedures slow the company's capacity to advertise new properties and reduce its market presence and competitiveness. In the real estate market, delays can be very costly. The present procedure also inhibits the company from optimising listing website advertising, resulting in incomplete or delayed property listings, less visibility, and decreased consumer engagement.

B. Justify the ML-based solution along with the requirements to solve the given problem.

Using artificial intelligence (AI) techniques to address business problems demonstrate the company's ability to utilise innovative technologies to remain at the forefront of the industry and demonstrate its commitment to innovation. This can be presented as a competitive advantage, appealing to prospective customers and setting the company apart. Automating the process of property image categorisation can have significant positive effects on the following aspects of the business:

1. **Time Efficiency**: By automating property image classification, staff could devote more time to creative and high-value tasks. Time efficiency improves employee productivity and operational effectiveness. Additionally, machine learning may increase employee loyalty by addressing pain points and facilitating meaningful work.
2. **Cost Reduction**: The company could save operational costs by decreasing its manual labour expenses. Staff can be better utilised, and manual labour can be eliminated, reducing the human resource costs associated with recruiting and training additional personnel to manage the increasing property volume. Additionally, automation frees up the employees' time for duties that generate revenue, thereby improving the company's finances.
3. **Employee Satisfaction**: By automating image classification, employees would be free to perform more innovative and significant work. The company can also improve employee morale, and job satisfaction increases retention and morale in the workplace. Employees who are engaged and satisfied are more likely to remain with the organisation and contribute to its success. The management team will be pleased

with the effective implementation of machine learning because it aligns with their goals of reducing human resource costs and enhancing operational efficiency.

4. **Scalability and Accuracy**: With an automated image classification system, the company can categorise the growing quantity of properties consistently and efficiently, enhancing property listing quality. This scalable and accurate technique helps to boost customer satisfaction and confidence. The company can eliminate manual errors (approximately 8 per cent now). Furthermore, this automated method enhances property image categorisation accuracy and consistency, resulting in higher-quality listings and a better company reputation. This improved user experience will attract consumers and renters to the company's services. Consequently, this improvement may boost client loyalty and repeat business.

C. Machine learning lifecycle (CRISP-MP) project proposal

1. Business Understanding: (see Appendix A for more information)

Topic	Key Information
Problem Statement (Business Objectives)	To deploy AI techniques to increase efficiency, employee satisfaction, and cost reduction in property photo uploading and categorisation.
Success Criteria (For Business)	• Reduction in labour costs • Increase work efficiency • Increase employee & customer satisfaction
Success Criteria (For Machine Learning)	• Accuracy of image categorisation • Speed of property listing
Legal and Compliance	The agents must obtain a written consent to publish the pictures on the websites, especially when they contain the owners' private possessions.
Proof of Concept (PoC)	**Business Value**: Implementing a machine learning model for automated photo classification can yield substantial benefits, including time and cost savings, enhanced

	employee satisfaction, improved accuracy and consistency, scalability, and a competitive edge. **Required Data**: Over 400,000 of property images in a month. **Time Frame:** Develop and deploy within 3 months of contract date.
Project Team and Roles	Head of Operation is the project manager. There are 10 people working in this project.
Budget and Resources	Total budget is $750,000 with all necessary resources supplied by ABC Consultancy
Stakeholder Engagement	Meeting at X company every 2 weeks. **Note:** Please minimise technical jargons or terminology as much as possible when communicating with the agency X.

2. **Data Understanding: (**see Appendix B for more information)

Description	Detail
Property Image Formats	JPEG, PNG, GIF
Image Collection	From all 3 branches and sent to the head office in New York
Dataset Size	Over 400,000 images monthly
Image Categories	Bedrooms, living rooms, bathrooms, kitchens, dining room, interior and exterior
Data Collection	Gathered photos from 4 separate folders from each branch.
Associated Metadata	Property type, location, estimated price, picture format, and size of picture
Data Quality	Routine data validation checks and visual inspection by staff, expert feedback and aim for images with a resolution of at least 1,024 x 1,024 pixels.
Impact on Model Performance	High-quality images lead to improved model performance and accuracy.

3. **Data Engineering:** This stage involves data pre-processing tasks such as data selection, cleaning, construction, and normalisation. In addition, the dataset will be split into 2 sets for this project. The training set is 80 per cent of the total dataset. It will be used to train

the machine learning model, and the testing set is 20 per cent which will be used to evaluate its performance (see Appendix C for more information).

Task	Details
Data Selection	Representative dataset covering diverse categories and property types.
	Sufficient examples in each category
	Balanced representation of each category
	Augmentation techniques to increase dataset diversity (if necessary)
Data Cleaning	Remove duplicates
	Exclude incomplete or corrupted images
	Evaluate image quality
	Remove images with noise, blurriness, etc.
Data Construction	Transform data into a suitable format for training the model
	Create or generate new data images
	Techniques used such as image orientation, cropping, resizing, and brightness/contrast adjustment
Data Normalisation	Transform data into a common scale/range
	Ensure fair comparisons
	Prevent dominance of specific characteristics in the analysis

4. Machine Learning Model Engineering:

- **Model Selection:** The classification accuracy, database size, image categorisation challenges, and budget determines the choice of machine learning model. The consulting team must also evaluate the project's viability regarding experience, computational resources, and infrastructure for model construction and deployment.

Initial consideration should be given to traditional supervised machine learning models, such as Decision Tree and the Random Forest model (Talari, 2022), that can extract relevant property image features and train classifiers on these features. However, the company should also consider more advanced and expensive machine learning models, such as the Inception model (Alake,2020).

Comparison of the models

Criteria	Decision Trees	Random Forests	Inception Model
Handling Data Types	Can handle both numerical and categorical data without requiring extensive pre-processing	Can handle both numerical and categorical data without requiring extensive pre-processing	Normally used for image classification tasks
Overfitting	Prone to overfitting, especially with deep and complex trees	Helps reducing overfitting through ensemble of trees and random feature selection	Prone to overfitting if not properly regularised or trained with large and diverse dataset
Performance	Can perform well with small to medium-sized dataset and not too complex problems	Generally, performs better than Decision Trees, especially with larger and more complex datasets	Achieves state of the art performance in image classification tasks and complex visual recognition problems
Computational Complexity	Low computational complexity	Higher computational complexity compared to Decision Trees	Higher computational complexity due to deep convolutional neural network architecture
Applicability	Suitable for both classification and regression tasks with small to medium-sized datasets	Suitable for both classification and regression tasks, especially with larger and complex datasets	Primary used for image classification tasks
Costs	Cheaper	Moderate	More expensive

- **Model Training:** The machine learning model is trained using 80 per cent of the data from the total dataset. The goal is to fine-tune the model and evaluate its performance.
- **Model Registration:** It is a crucial stage in the machine learning development lifecycle, at this stage of the machine learning development lifecycle, the trained model's version, configuration, hyperparameters, training data, evaluation metrics, and metadata must be collected and saved for future deployment.

5. **Model Evaluation and Deployment:**

- **Model Evaluation**: Various metrics must be aligned with the objectives and requirements of the project when evaluating the machine learning model for automated image classification. These metrics will comprehensively evaluate the model's efficiency and performance.

  Therefore, we will prioritise these metrics based on the company's requirements, considering the significance of accuracy, precision, explainability, robustness, and latency.

  There are 2 categories that comprise the evaluation: performance metrics and soft measures (see Appendix D for more information).

Performance Metrics	Details
Accuracy (weight 40 per cent)	Target 98 per cent or higher
	Measures the percentage of correctly categorised photos from the total dataset.
	Overfitting is detected if accuracy drops significantly in the testing stage.
Precision (weight 10 per cent)	Target 98 per cent or higher
	Measures accuracy of positive case identification.
	Indicates accuracy in labelling correctly.
Recall Rate (weight 10 per cent)	Target 98 per cent or higher

	Demonstrates the model's ability to avoid mistakes and capture all relevant features for each category.
	Ensures accuracy and completeness of automated photo classification.
Soft Measures	**Details**
Explainability (weight 10 per cent)	Understanding of the model's image classification process in order to obtain insight into which image features contribute most to the model's decisions.
Robustness (weight 20 per cent)	Ability to adapt to image variations and challenges.
	Reliability in real-world scenarios.
Latency (weight 10 per cent)	The length of time to analyse and categorise a property images.
	Real-time or near-real-time performance
	Meets acceptable timeframes for effective automation.

- **Model Deployment**: The company will use the Cloud services offered by ABC Consultancy and this service is already included in the total budget. Furthermore, the team will deploy **the Canary Deployment strategy** to safely roll out a new model. Before replacing the current manual process, it introduces the new model to a portion of the workload or user base for extensive testing and validation in real-world circumstances. This regulated and methodical approach minimises disruption, improves adoption of the new model, and allows for early problem-solving, improving risk reduction and validation before committing to the new system (Awati and Loshin, 2022).

6. **Machine Learning Model Monitoring:** The project team will have to: -

- Continuously monitor and evaluate the accuracy and reliability of the model's performance in the real-world scenario.

- Collect user feedback and establish explicit communication channels to gather user feedback and stakeholder requirements.
- Periodically evaluate the model's performance and retrain if required to maintain high accuracy.
- Update the training dataset and retrain the model to enhance performance as new property images and categories emerge.
- Implement a process for version management and regular updates.
- Monitor the quality of input data in order to identify potential issues, such as incorrect or biased labels.

D. Propose MLOps design for development and operation.

Organisations should begin with the Manual Process (Maturity Level 0), where machine operations are performed manually and progress through the higher maturity levels as they implement and administer machine learning models in production.

Level 2 (Maturity Level 2) machine learning operations or an automated Continuous Integration (CI) and Continuous Deployment (CD) system should be adopted by the company. Level 2 automates machine learning models' training, evaluation, and deployment in a continuous integration and deployment pipeline (Reshytko, 2022). Using this website-integrated pipeline, employees can rapidly submit property images for classification.

These infrastructure phases automate the formulation and deployment of machine learning models. This will increase the efficacy, effectiveness, and dependability of advertising campaigns.

1. **Source Control Management:** A centralised source control repository will be established (in New York office) to contain all project-related code, configuration

files, and resources. This will ensure version control, collaboration, and reproducibility, enabling multiple team members to work simultaneously on the tasks.

2. **Continuous Integration:** We will establish a continuous integration stage in which various developers from ABC Consultant can routinely modify the changes and merge into the shared repository. This process will compile the code and perform any required pre-processing or data transformations. Continuous integration ensures the seamless incorporation of code changes and the early detection of integration issues.
3. **Automated Testing:** Once the code has been developed, the pipeline will transition to the automated testing phase. Automated testing will guarantee that the model satisfies the predefined quality standards and performs as expected.
4. **Artifact Generation**: The pipeline will generate artifacts (or the output from the training process) necessary for deployment after effective testing. These artifacts include the trained machine learning model and other necessary files. The generated artifacts will be saved and ready for deployment in the subsequent phase.
5. **Deployment:** The generated artifacts will be deployed to the production environment. This procedure will be automated to ensure consistency and dependability. It will be integrated with the existing website platforms. Automation will allow for efficient and scalable deployment, reducing the manual labour required for the process.
6. **Monitoring and Feedback Loop**: Once the model has been deployed, we will implement a monitoring system to perpetually track its performance in the production environment. We will capture metrics such as accuracy, latency, and resource consumption to evaluate the model's behaviour. This monitoring will detect any problems or anomalies quickly, activating alerts or automated actions as required. The feedback cycle between the monitoring and development phases will continuously enhance the model's performance.

By implementing Maturity Level 2, machine learning operations will provide the organisation with the automation, scalability, and dependability necessary to speed up the process of uploading property images to the listing website portals.

Appendix A

- **Problem statement:** The company seeks to deploy AI techniques to increase efficiency, increase employee satisfaction, and reduce human resource costs from the current manual categorisation and uploading of property photos.
- **Success Criteria:** To evaluate the success of the project the following Key Performance Indicators (KPIs) can be considered. However, the success of the solution will be measured by 2 different criteria.

1. **Business Success Criteria**
   - **Reduction in Labour Costs**: Calculate the cost savings resulting from reducing the time of manual photo categorisation labour hours. This KPI measures the financial impact of automation by quantifying the reduction in labour costs associated with a given task. For example, the company only require having 1 person to do this task rather than 4 people. Therefore, the company can save $59,280 * 3 = $177,840 on annual salary.
   - **Employee Satisfaction**: Collect feedback from employees who participated in the photo categorisation process regarding their impressions of the automated system. This KPI evaluates the machine learning solution's impact on employee satisfaction, engagement, and productivity.

- **Customer Satisfaction**: Collect customers' feedback regarding their experience with the property listings and the photo categorisation process. This can be accomplished via surveys or reviews. This KPI measures the level of consumer satisfaction and provides insight into how well the automated system met the company's expectations. Positive feedbacks indicate efficiency gain from automated system. For example, the target of positive feedback from the survey is more than 80 per cent within 3 months of implementing this new system.
- **Cost-benefit Analysis**: Conduct a cost-benefit analysis to determine the project's financial impact. This includes comparing the machine learning system's initial investment and ongoing maintenance costs to the cost savings, efficiency increases, and revenue generated. This KPI helps determine a project's return on investment (ROI) and overall financial success.

2. Machine Learning Success Criteria

- **Accuracy of Image Categorisation**: This KPI measures the accuracy of an automated process assigning images to predefined categories, such as bedroom, living room, and kitchen. For example, the target of accuracy is 98 per cent or higher. The company can compare the accuracy between manual and automated categorisation.
- **Speed of Property Listings**: Measure the time between acquiring property and its listing on the website. This KPI measures the effectiveness of the automated process in reducing “time to market” by expediting the launch of new properties on the website. For example, the target of “time to market” is 2 days.

- **Legal and Compliance:** The agent must obtain a written consent to publish the pictures on the website, especially when they contain the owners' private possessions.
- **Proof of Concept (PoC) and Product Development Plan**: The PoC aims to validate the technical viability requirements, accuracy, and potential benefits of the machine learning solution to mitigate risks of failures.
    - **Business Value:** Implementing a machine learning model to automate the company's photo classification process can result in significant business benefits, such as time and cost savings, increased employee satisfaction, improved accuracy and consistency, scalability, and competitive advantage.
    - **Required data:** Over 400,000 property images combined every month from the company's databases from all 4 branches.
    - **Time frame:** The consulting team aims to develop and deploy the machine learning model within 3 months from the contract date.
    - **Project team and roles:**
        1. Head of Operation: - Tim Cook
        2. Chief Financial Officer: - Tom Hank
        3. Head of the engineering team: - Lisa May
        4. Data Engineer: - Tom Jones
        5. Data Scientist: - James Cameron
        6. Web Developer: - Mary Nars
        7. The administrators in each branch (who are manually performing this task now)
    - **Budget and resources:** The total budget for this project is $750,000. ABC Consultancy will supply all the resources needed to complete the project,

including maintenance and updates, with no extra costs. ABC Consultancy has 5 years guaranteed policy.

- **Stakeholder engagement:** The project team will have a meeting every 2 weeks to update the progress of the project.

Appendix B

Property images could be in different **formats**, such as JPEG, PNG or GIF, and they are collected from all 4 branches and sent to the head office in New York. This large dataset covers various categories such as bedrooms, living rooms, bathrooms, and kitchens. The staff responsible for this task will have to collect a considerable amount of data (over 400,000 images monthly) from 4 separate folders from each branch. Along with photos, the associated **metadata** for each property, such as property type, location, estimated price, format and size of the picture, is required. Furthermore, **data quality** is essential for effectively training a machine learning model. The responsible staff must conduct routine **data validation** checks, visually inspect the images, and solicit expert feedback to identify and resolve data issues or inconsistencies. Images of superior quality will contribute to improved model performance and more accurate predictions.

Appendix C

- **Data Selection:** This task involves gathering property photos from the company's database from each branch. The company has to select a representative dataset of property images that cover a diverse range of categories and property types. In addition, there must be sufficient examples of images in each category and ensure balanced representation. Moreover, augmentation techniques could be used to increase the diversity of the dataset.

- **Data Cleaning:** Data quality is crucial for effective training, and cleansing image datasets involves assuring the images' consistency and quality and avoiding bias. This may involve removing duplicates, excluding incomplete or corrupted images, evaluating image quality, and removing images with significant noise, blurriness, or artefacts that may hinder the model's performance.
- **Data Construction and Normalisation:** This task involves transforming data into a suitable format for training the model. This may include creating or generating new data images based on an existing dataset (or data augmentation) utilises various techniques such as image orientation, cropping images, resizing the image, and adjusting the brightness and contrast. In addition, data normalisation could be performed to transform data into a common scale or range in order to ensure fair comparisons and prevent particular characteristics from dominating the analysis.

Appendix D

1. Performance metrics: -

- **Accuracy of 98 per cent or higher:** This metric measures the percentage of property photos correctly categorised out of the total number of photos. It demonstrates the broad correctness of the model's predictions. For example, if the model's accuracy is 95 per cent in the training stage but only 75 per cent in the testing stage, it means that the model is overfitting.
- **Precision of 98 per cent or higher:** It measures the model's ability to accurately identify positive cases, such as correctly classifying a photo as a bedroom, out of all

instances predicted as positive. In addition, it indicates the model's accuracy in correctly designating labels.

- **Recall rate of 98 per cent or higher:** It demonstrates the model's ability to avoid mistakes and capture all relevant features for each category. By evaluating the model's recall for each category, the team can evaluate its ability to capture and classify the relevant features associated with different property photo categories, thereby ensuring the accuracy and completeness of the automated photo classification process.

2. Soft measures: -

- **Explainability:** It is essential to understand how the model classifies images in order to obtain insight into which image features contribute most to the model's decisions. This will aid our understanding of the model's decision-making process, eliminating bias, and increase its transparency.
- **Robustness:** It refers to the model's ability to adapt to variations in property image quality, lighting conditions, and other potential challenges. This will ensure the model's reliability in real-world situations.
- **Latency:** It measures how long the model takes to analyse and categorise a property image. Considering hardware and infrastructure limitations, the team will determine if the model can provide real-time or near-real-time performance. This will ensure that the model's predictions can be generated within acceptable timeframes for effective automation.

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